P-polarized lattice sum

This set of notes concerns the evaluation of the lattice sum(s) required to obtain the scattering properties of p-polarized waves off of a one-dimensional grating of infinitely long cylinders (see e.g. **Equations 20 and 21** of the paper Extraordinary optical reflection from sub-wavelength cylinder arrays (https://opg.optica.org/directpdfaccess/275739ab-ef21-4be2-89d7a52466f2c6b9_89577/oe-14-9-3730.pdf?da=1&id=89577&seq=0&mobile=no) by **Gomez-Medina et al** (henceforth referred to as **Paper 1**)). The crux of the derivation is contained in **Equations 35-37** of the paper Dynamic Interaction Fields in a Two Dimensional Lattice (https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=1125284) by **Collin et al** (henceforth referred to as **Paper 2**). Our approach below is equivalent to the approach of **Paper 2**, though we differ slightly in methodology. In short, we wish to calculate the following ***restricted lattice sum***:

$$\sum_{m=-\infty}^{\infty} e^{iQ_0 x_m} \partial_y^2 G(|\mathbf{r}_0 - \mathbf{r}_m|, k_0) - \lim_{\mathbf{r}\to\mathbf{r}_0} \partial_y^2 G(|\mathbf{r} - \mathbf{r}_0|, k_0),$$

which determines the interaction of the lattice site at the origin with all other lattice sites. We note that $G(r, k_0) \equiv \frac{i}{4} H_0^{(1)}(k_0 r)$ is the ***non-interacting*** Green's function of the Helmholtz equation in two dimensions with $k_0 \equiv \omega/c$ being the wavenumber in free space and $H_0^{(1)}$ being the Hankel function of the first kind (with outgoing boundary conditions) of order zero. In the two sections, below, we calculate the restricted lattice sum by computing its two consituent parts:

$$\sum_{m=-\infty}^{\infty} e^{iQ_0 x_m} \partial_y^2 G(|\mathbf{r}_0 - \mathbf{r}_m|, k_0),$$

which we term the ***unrestricted lattice sum*** (evaluated in **Section 1**) and

$$\lim_{\mathbf{r}\to\mathbf{r}_0} \partial_y^2 G(|\mathbf{r} - \mathbf{r}_0|, k_0),$$

which we denote the ***self-energy term*** (evaluated in **Section 2**).

Section 1: The unrestricted lattice sum

We write the unrestricted lattice sum, as defined above, as follows:

$$\lim_{\mathbf{r}\to\mathbf{r}_0}\sum_{m=-\infty}^{\infty} e^{iQ_0x_m}\partial_y^2 G(|\mathbf{r}-\mathbf{r}_m|,k_0) = \lim_{x\to 0,y\to 0} -\frac{i}{2D}\sum_{m=-\infty}^{\infty} q_m e^{i(Q_0-K_m)x}e^{iq_m|y|},$$

with $q_m \equiv \sqrt{k_0^2-(Q_0-K_m)^2}$ (using the notation of **Paper 1**) and where we used the following:

$$\sum_{m=-\infty}^{\infty} e^{iQ_0x_m}G(|\mathbf{r}-\mathbf{r}_m|,k_0) = \frac{1}{4\pi^2}\int\int\sum_{m=-\infty}^{\infty}\frac{e^{iQ_0x_m}e^{ik_x(x-x_m)}e^{ik_yy}}{k^2-k_0^2}\mathrm{d}k_x\mathrm{d}k_y =$$
$$\frac{1}{4\pi^2}\frac{2\pi}{D}\int\sum_{m=-\infty}^{\infty}\frac{e^{i(Q_0-K_m)x}e^{ik_yy}}{(Q_0-K_m)^2+k_y^2-k_0^2}\mathrm{d}k_y = \frac{i}{2D}\sum_{m=-\infty}^{\infty}\frac{e^{i(Q_0-K_m)x}e^{iq_m|y|}}{q_m},$$

where the last equality follows from contour integration. Therefore, the unrestricted lattice sum we wish to calculate may be written as follows:

$$\lim_{x\to 0,y\to 0} -\frac{i}{2D}\sum_{m=-\infty}^{\infty} q_m e^{i(Q_0-K_m)x}e^{iq_m|y|}$$

Evaluated at $x=y=0$, the result is simple and gives us the following:

$$-\frac{1}{2D}\sum_{m=1}^{\infty}\left[iq_m+iq_{-m}\right]-\frac{iq_0}{2D}$$

This accounts for three of the terms in **Equation 21** of **Paper 1**.

Section 2: Evaluation of the self-energy term

We now calculate the self-energy term, which must be subtracted from the unrestricted lattice sum in order to evaluate the restricted lattice sum.:

$$\frac{i}{4}\partial_y^2 H_0^{(1)}(k_0r) = -\frac{ik_0}{4}\partial_y\left[\frac{y}{r}H_1^{(1)}(k_0r)\right] = -\frac{ik_0}{4}\left[\frac{1}{r}H_1^{(1)}(k_0r)-\frac{y^2}{r^3}H_1^{(1)}(k_0r)+\frac{k_0y^2}{r^2}H_1^{(1)}(k_0r)'\right],$$

where the prime in the last term denotes differentiation with respect to the argument, k_0r. We set $x=0$ and take the limit $y\to 0$, which gives us the following:

$$\frac{i}{4}\partial_y^2 H_0^{(1)}(k_0|y|) = -\frac{ik_0^2}{4}\left[H_0^{(1)}(k_0|y|)-\frac{1}{k_0|y|}H_1^{(1)}(k_0|y|)\right] =$$
$$= -\frac{ik_0^2}{4}\left[J_0(k_0|y|)-\frac{1}{k_0|y|}J_1(k_0|y|)\right]+\frac{k_0^2}{4}\left[Y_0(k_0|y|)-\frac{1}{k_0|y|}Y_1(k_0|y|)\right]$$

The first term in the above, which has only Bessel functions of the first kind is easy. For the second term, we use the expansion of $Y_0(z)$ given by Wolfram (link: [Bessel Function of the Second Kind](https://mathworld.wolfram.com/BesselFunctionoftheSecondKind.html) (https://mathworld.wolfram.com/BesselFunctionoftheSecondKind.html)):

$$Y_0(z) \approx \frac{2}{\pi}\left[\ln(z/2)+\gamma\right]J_0(z)+\frac{2}{\pi}\frac{z^2}{4} \to Y_1(z) = -Y_0'(z) \approx -\frac{2}{\pi}\left[\frac{1}{z}J_0(z)+J_0'(z)\Big[\ln(z/2)+\gamma\Big]+\frac{z}{2}\right]$$
$$\approx -\frac{2}{\pi}\left[\frac{1}{z}+\frac{z}{4}-\frac{z}{2}\ln(z/2)-\frac{z}{2}\gamma\right]$$

Therefore, we obtain:

$$Y_0(k_0|y|)-\frac{1}{k_0|y|}Y_1(k_0|y|) \approx \frac{2}{\pi}\left[\frac{1}{2}\ln\left[\frac{k_0|y|}{2}\right]+\frac{\gamma}{2}+\frac{1}{4}+\frac{1}{k_0^2y^2}\right],$$

which allows us to write the self-energy term as follows:

$$\frac{i}{4}\partial_y^2 H_0^{(1)}(k_0|y|) \approx -\frac{ik_0^2}{8}+\frac{k_0^2}{4\pi}\left[\ln\left[\frac{k_0|y|}{2}\right]+\gamma+\frac{1}{2}+\frac{2}{(k_0y)^2}\right]$$

Note that this clearly has logarithmically and quadratically divergent terms which we must address. To address the logarithmic term, we write:

$$\ln\left[\frac{k_0|y|}{2}\right] = \ln\left[\frac{k_0D}{4\pi}\frac{2\pi|y|}{D}\right] = \ln\left[\frac{k_0D}{4\pi}\right]+\ln\left[\frac{2\pi|y|}{D}\right] = \ln\left[\frac{k_0D}{4\pi}\right]-\frac{2\pi}{D}\sum_{m=1}^{\infty}\frac{D}{2\pi m},$$

where the second equality follows from our previous notes (see [Coupled dipole model for coherent perfect absorption](https://hackmd.io/@aligho/HJXva_Dv-e) (https://hackmd.io/@aligho/HJXva_Dv-e)) and it should be understood that we are taking the $y \to 0$ limit. Furthermore, to tackle the quadratically divergent term, we write:

$$\sum_{m=1}^{\infty} m = \lim_{y\to 0}\sum_{m=1}^{\infty} m e^{-2\pi m|y|/D} = -\frac{D}{2\pi}\lim_{y\to 0^+}\partial_y\sum_{m=1}^{\infty}e^{-2\pi m y/D} = -\lim_{y\to 0^+}\frac{D}{2\pi}\partial_y\frac{e^{-2\pi y/D}}{1-e^{-2\pi y/D}} =$$
$$\lim_{y\to 0^+}\left[\frac{e^{-2\pi y/D}(1-e^{-2\pi y/D})+e^{-2\pi y/D}e^{-2\pi y/D}}{(1-e^{-2\pi y/D})^2}\right] = \lim_{y\to 0^+}\frac{1}{(e^{\pi y/D}-e^{-\pi y/D})^2} \approx$$
$$\approx \lim_{y\to 0^+}\frac{1}{(2\pi y/D)^2[1+(\pi y/D)^2/6]^2} \approx \lim_{y\to 0}\frac{D^2}{4\pi^2y^2}\left[1-\frac{\pi^2y^2}{3D^2}\right]$$

From this, we obtain the following:

$$\lim_{y\to 0}\frac{1}{2\pi y^2}=\frac{2\pi}{D^2}\sum_{m=1}^{\infty}m+\frac{\pi}{6D^2}$$

With all of these ingredients together, we may write the self-energy term as follows:

$$-\frac{ik_0^2}{8}+\frac{k_0^2}{4\pi}\left[\ln\frac{k_0 D}{4\pi}+\gamma+\frac{1}{2}\right]+\frac{\pi}{6D^2}-\frac{1}{D}\sum_{m=1}^{\infty}\left[-\frac{2\pi m}{D}+\frac{k_0^2}{2}\frac{D}{2\pi m}\right]$$

Subtracting this from the ***restricted lattice sum*** evaluated in **Section 1** gives us **Equation 21** of **Paper 1**.